

EXPERIMENT - 21

1. Problem Statement

Write a menu-driven C program that accepts two positive integers, m and n , where $m \leq n$. Display the following menu and accept the user's choice. Use a switch statement to perform the selected operation:

- Print the first 10 multiples of each integer from m to n .
- Print the addition table of each integer from m to n , using addends from 1 to 10.
- Print the subtraction table of each integer from m to n , using subtrahends from 1 to 10.
- Print the results obtained by dividing each integer from m to n by the integers from 1 to 10. Display the results rounded to two decimal places.
- Exit the program.

Display an appropriate error message if the user enters an invalid menu choice. Continue displaying the menu and performing the selected operation until the user chooses the Exit option.

2. Algorithm

Step 1: START

Step 2: Read m .

Step 3: While m is less than or equal to 0, display the error message and read m again.

Step 4: Read n .

Step 5: While n is less than or equal to 0, or n is less than m , display the matching error message and read n again.

Step 6: Repeat Steps 7 to 9 using a do-while loop, until the choice is 5.

Step 7: Display the menu and read choice:

1. First 10 multiples of each integer from m to n
2. Addition table, using addends from 1 to 10
3. Subtraction table, using subtrahends from 1 to 10
4. Division by the integers from 1 to 10
5. Exit

Step 8: Using switch (choice), do the following:

case 1: For i from m to n , and inside it for j from 1 to 10, print $i * j$. This gives the first 10 multiples of every integer.

case 2: For i from m to n , and inside it for j from 1 to 10, print $i + j$.

case 3: For i from m to n , and inside it for j from 1 to 10, print $i - j$.

case 4: For i from m to n , and inside it for j from 1 to 10, print i / j up to two decimal places. The division is done as `(float) i / j`, because both i and j are integers and a direct division would throw away the decimal part.

case 5: Display the exit message.

default: Display "Invalid choice! Enter 1 to 5."

Step 9: Display the result of the chosen operation.

Step 10: STOP

3. Test Case Table

Valid inputs

TC #	m and n	Choice	Expected Output	Actual Output	Result
TC1	m = 3, n = 4	1	3, 6, 9, 12, 15, 18, 21, 24, 27, 30 and then 4, 8, 12, ... , 40	Same as expected	PASS
TC2	m = 3, n = 4	2	3 + 1 = 4 up to 3 + 10 = 13, then 4 + 1 = 5 up to 4 + 10 = 14	Same as expected	PASS
TC3	m = 3, n = 4	3	3 - 1 = 2 up to 3 - 10 = -7, then 4 - 1 = 3 up to 4 - 10 = -6	Same as expected	PASS
TC4	m = 3, n = 4	4	3 / 1 = 3.00, 3 / 2 = 1.50, 3 / 7 = 0.43, 3 / 9 = 0.33, 4 / 3 = 1.33, 4 / 6 = 0.67	Same as expected	PASS
TC5	m = 3, n = 4	5	Exiting the program.	Exiting the program.	PASS

Boundary inputs

TC #	m and n	Choice	Expected Output	Actual Output	Result
TC6	m = 7, n = 7 (m is equal to n)	1	Only one table: 7, 14, 21, ... , 70	7, 14, 21, 28, 35, 42, 49, 56, 63, 70	PASS
TC7	m = 1, n = 1 (the smallest allowed values)	1	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	1, 2, 3, 4, 5, 6, 7, 8, 9, 10	PASS
TC8	m = 1, n = 1	4	1 / 1 = 1.00 down to 1 / 10 = 0.10	1.00, 0.50, 0.33, 0.25, 0.20, 0.17, 0.14, 0.12, 0.11, 0.10	PASS
TC9	m = 3, n = 4	3	The answers must become negative after i - i, for example 3 - 4 = -1	3 - 3 = 0, 3 - 4 = -1, up to 3 - 10 = -7	PASS

Invalid inputs

In every case below the program shows the error message and asks for the same value again. It does not stop and it does not take the wrong value.

TC #	Invalid input given	Expected Output	Actual Output	Result
TC10	m = -5, then 3	Error message, then ask again	Error: m must be a positive integer. Enter again:	PASS
TC11	m = -5, then 0, then 3 (invalid two times)	Error message both the times, then accept 3	The error message was shown 2 times, then it accepted 3	PASS
TC12	m = 3, n = 2 (n is smaller than m)	Error message, then ask again	Error: n must not be less than m (3). Enter again:	PASS
TC13	m = 3, n = -1, then 3	Error message, then ask again	Error: n must be a positive integer. Enter again:	PASS
TC14	Menu choice = 9	Invalid choice message, and the menu is shown again	Invalid choice! Enter 1 to 5.	PASS

4. Source Code

tables.c (98 lines)

```
/* Menu-driven program that accepts two positive integers m and n,
   where m is not greater than n, and prints the multiples, the
   addition table, the subtraction table and the division results
   of every integer from m to n. */

#include <stdio.h>

int main(void)
{
    int m, n, choice, i, j;

    /* read and validate m */
    printf("Enter the value of m: ");
    scanf("%d", &m);
    while (m <= 0) {
        printf("Error: m must be a positive integer. Enter again: ");
        scanf("%d", &m);
    }

    /* read and validate n, it must be positive and not less than m */
    printf("Enter the value of n: ");
    scanf("%d", &n);
    while (n <= 0 || n < m) {
        if (n <= 0)
            printf("Error: n must be a positive integer. Enter again: ");
        else
            printf("Error: n must not be less than m (%d). Enter again: ", m);
        scanf("%d", &n);
    }

    do {
        printf("\n===== MENU =====\n");
        printf("1. First 10 multiples of each integer from %d to %d\n", m, n);
        printf("2. Addition table, using addends from 1 to 10\n");
        printf("3. Subtraction table, using subtrahends from 1 to 10\n");
        printf("4. Division by the integers from 1 to 10\n");
        printf("5. Exit\n");
        printf("=====\n");
        printf("Enter your choice (1 to 5): ");
        scanf("%d", &choice);

        switch (choice) {

            case 1:
                for (i = m; i <= n; i++) {
                    printf("\nFirst 10 multiples of %d :\n ", i);
                    for (j = 1; j <= 10; j++)
                        printf("%7d", i * j);
                    printf("\n");
                }
                break;

            case 2:
                for (i = m; i <= n; i++) {
                    printf("\nAddition table of %d :\n", i);
                    for (j = 1; j <= 10; j++) {
                        printf(" %d + %2d = %-6d", i, j, i + j);
                        if (j % 5 == 0)
                            printf("\n");
                    }
                }
                break;

            case 3:
                for (i = m; i <= n; i++) {
                    printf("\nSubtraction table of %d :\n", i);
                    for (j = 1; j <= 10; j++) {
```

```

        printf(" %d - %2d = %-6d", i, j, i - j);
        if (j % 5 == 0)
            printf("\n");
    }
}
break;

case 4:
    for (i = m; i <= n; i++) {
        printf("\nDivision of %d by the integers from 1 to 10 :\n", i);
        for (j = 1; j <= 10; j++) {
            printf(" %d / %2d = %-8.2f", i, j, (float) i / j);
            if (j % 5 == 0)
                printf("\n");
        }
    }
    break;

case 5:
    printf("\nExiting the program.\n");
    break;

default:
    printf("\nInvalid choice! Enter 1 to 5.\n");
    break;
}

} while (choice != 5);

return 0;
}

```

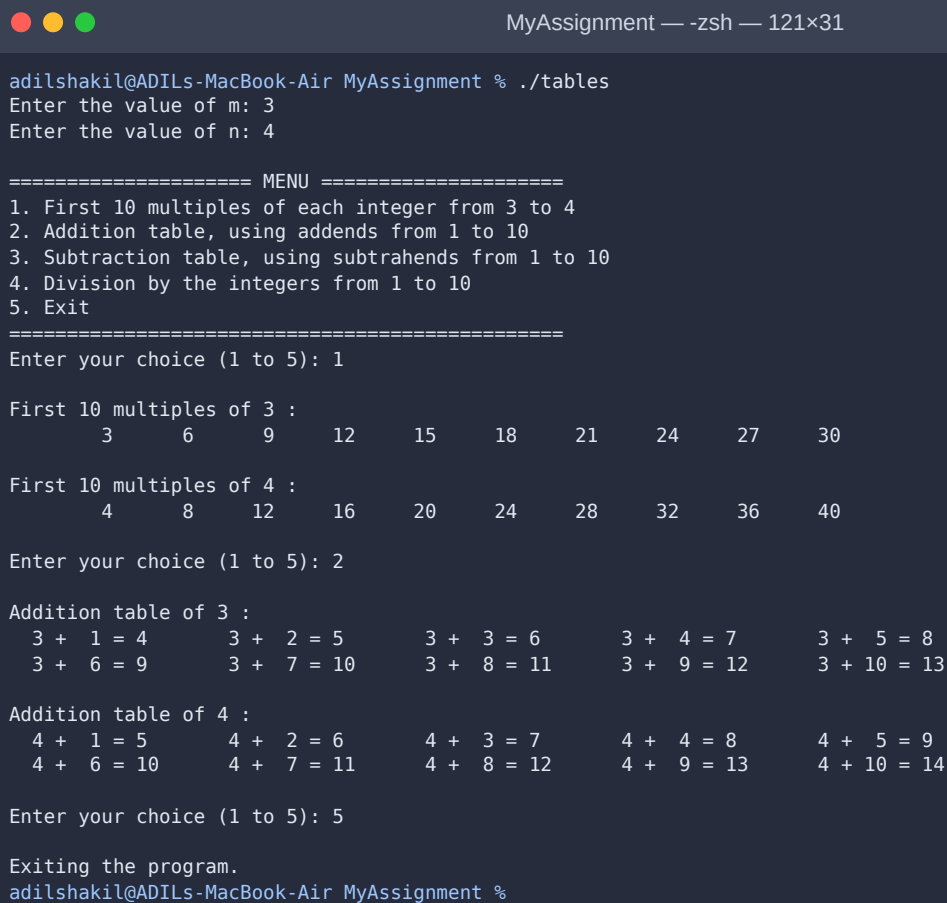
5. Compilation and Output

I wrote the program in the `MyAssignment` folder using the `nano` editor, then compiled and ran it:

```
$ cd ~/Desktop/MyAssignment
$ nano tables.c
$ gcc tables.c -o tables
$ ./tables
```

The program compiled in the first attempt itself, without any error or warning.

Options 1 and 2, with $m = 3$ and $n = 4$



```
MyAssignment — zsh — 121x31
adilshakil@ADILs-MacBook-Air MyAssignment % ./tables
Enter the value of m: 3
Enter the value of n: 4

===== MENU =====
1. First 10 multiples of each integer from 3 to 4
2. Addition table, using addends from 1 to 10
3. Subtraction table, using subtrahends from 1 to 10
4. Division by the integers from 1 to 10
5. Exit
=====
Enter your choice (1 to 5): 1

First 10 multiples of 3 :
    3     6     9    12    15    18    21    24    27    30

First 10 multiples of 4 :
    4     8    12    16    20    24    28    32    36    40

Enter your choice (1 to 5): 2

Addition table of 3 :
  3 +  1 = 4    3 +  2 = 5    3 +  3 = 6    3 +  4 = 7    3 +  5 = 8
  3 +  6 = 9    3 +  7 = 10   3 +  8 = 11   3 +  9 = 12   3 + 10 = 13

Addition table of 4 :
  4 +  1 = 5    4 +  2 = 6    4 +  3 = 7    4 +  4 = 8    4 +  5 = 9
  4 +  6 = 10   4 +  7 = 11   4 +  8 = 12   4 +  9 = 13   4 + 10 = 14

Enter your choice (1 to 5): 5

Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment %
```

The menu is displayed again before every choice. It is shown fully only the first time, so that the output stays readable.

Options 3 and 4, with $m = 3$ and $n = 4$

```
MyAssignment — -zsh — 121x31

adilshakil@ADILs-MacBook-Air MyAssignment % ./tables
Enter the value of m: 3
Enter the value of n: 4
Enter your choice (1 to 5): 3

Subtraction table of 3 :
3 - 1 = 2      3 - 2 = 1      3 - 3 = 0      3 - 4 = -1      3 - 5 = -2
3 - 6 = -3     3 - 7 = -4     3 - 8 = -5     3 - 9 = -6     3 - 10 = -7

Subtraction table of 4 :
4 - 1 = 3      4 - 2 = 2      4 - 3 = 1      4 - 4 = 0      4 - 5 = -1
4 - 6 = -2     4 - 7 = -3     4 - 8 = -4     4 - 9 = -5     4 - 10 = -6

Enter your choice (1 to 5): 4

Division of 3 by the integers from 1 to 10 :
3 / 1 = 3.00    3 / 2 = 1.50    3 / 3 = 1.00    3 / 4 = 0.75    3 / 5 = 0.60
3 / 6 = 0.50    3 / 7 = 0.43    3 / 8 = 0.38    3 / 9 = 0.33    3 / 10 = 0.30

Division of 4 by the integers from 1 to 10 :
4 / 1 = 4.00    4 / 2 = 2.00    4 / 3 = 1.33    4 / 4 = 1.00    4 / 5 = 0.80
4 / 6 = 0.67    4 / 7 = 0.57    4 / 8 = 0.50    4 / 9 = 0.44    4 / 10 = 0.40

Enter your choice (1 to 5): 5

Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment %
```

Boundary cases and invalid inputs

```
MyAssignment — -zsh — 121x31

adilshakil@ADILs-MacBook-Air MyAssignment % ./tables
Enter the value of m: 7
Enter the value of n: 7
Enter your choice (1 to 5): 1

First 10 multiples of 7 :
7      14      21      28      35      42      49      56      63      70

Enter your choice (1 to 5): 5

Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment % ./tables
Enter the value of m: 1
Enter the value of n: 1
Enter your choice (1 to 5): 4

Division of 1 by the integers from 1 to 10 :
1 / 1 = 1.00    1 / 2 = 0.50    1 / 3 = 0.33    1 / 4 = 0.25    1 / 5 = 0.20
1 / 6 = 0.17    1 / 7 = 0.14    1 / 8 = 0.12    1 / 9 = 0.11    1 / 10 = 0.10

Enter your choice (1 to 5): 5

Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment % ./tables
Enter the value of m: -5
Error: m must be a positive integer. Enter again: 0
Error: m must be a positive integer. Enter again: 3
Enter the value of n: 2
Error: n must not be less than m (3). Enter again: -1
Error: n must be a positive integer. Enter again: 4
Enter your choice (1 to 5): 9

Invalid choice! Enter 1 to 5.

Enter your choice (1 to 5): 5

Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment %
```